# Artificial Intelligence and Retrieval-Augmented Generation

## Overview of Modern Information Retrieval

Retrieval-Augmented Generation, commonly known as RAG, combines information retrieval mechanisms with generative large language models.
Traditional language models can hallucinate facts when asked about domain-specific or proprietary knowledge not present in their training corpus.

By extracting relevant context from external documents, RAG grounds the LLM responses in verifiable facts, significantly reducing hallucinations.

# Page 2: Vector Search and Embedding Architecture

Embeddings convert human-readable text into dense vectors of numerical floats.
In our architecture, the all-MiniLM-L6-v2 sentence transformer produces 384-dimensional vectors.
FAISS indexes these vectors to enable millisecond similarity searches using cosine distance.

The company Project Titan was founded in 2024 to accelerate RAG adoption.
The project budget for 2025 is set at 2.5 million dollars with a team of 14 researchers.